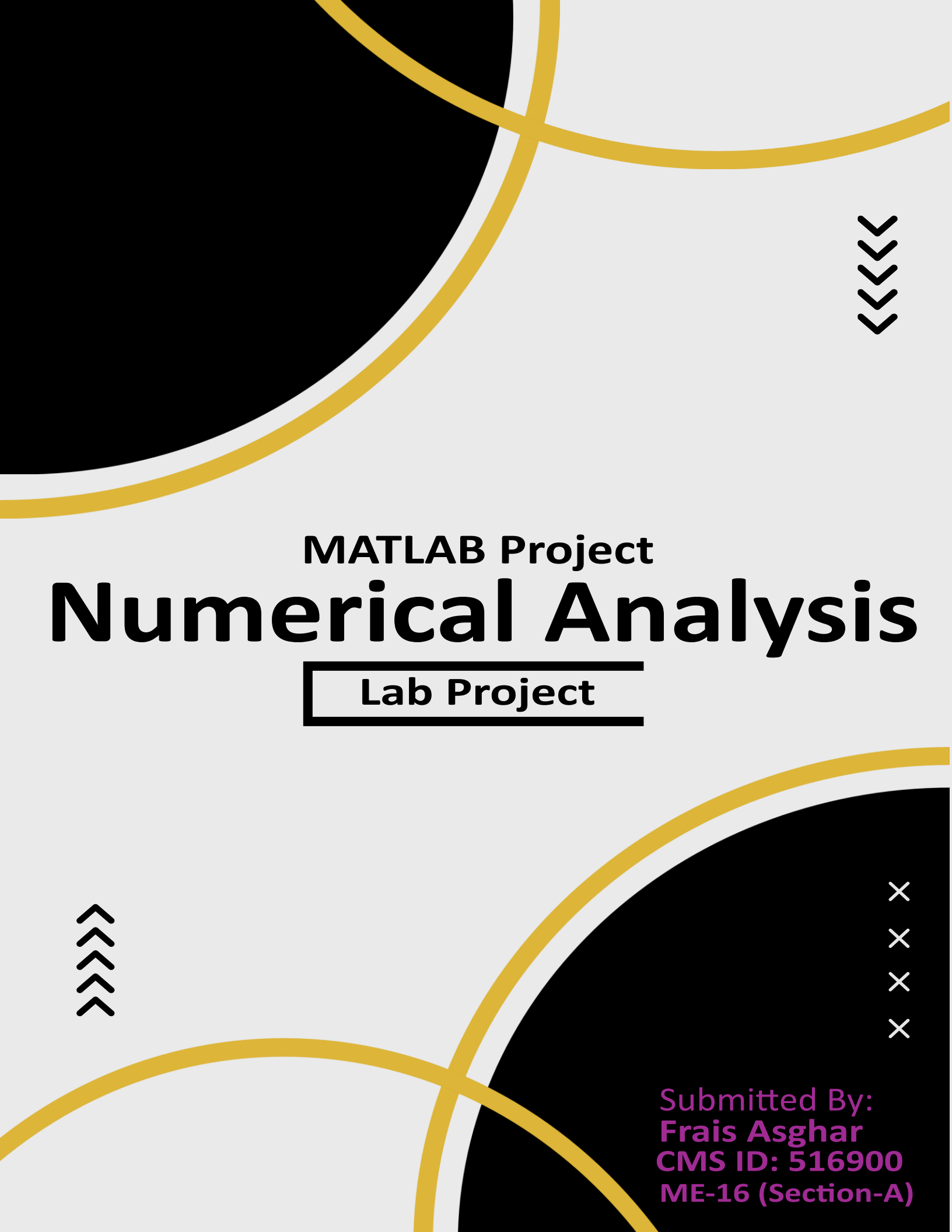




MATLAB Project

Numerical Analysis

Lab Project



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ME-16 (Section-A)



1. Introduction

Interpolation is a fundamental numerical technique used to estimate unknown values between known data points. In engineering and science, experimental data is typically obtained at discrete measurement points, and interpolation bridges the gaps to provide continuous estimates.

This project applies four interpolation methods to a set of experimental stress-strain data representing the mechanical behavior of a material. The objectives are to:

- ❖ Implement Lagrange interpolation from first principles.
- ❖ Implement Newton's Divided Difference interpolation and display the full divided-difference table.
- ❖ Apply MATLAB's built-in `interp1` function for linear interpolation.
- ❖ Apply cubic spline interpolation for smooth curve fitting.
- ❖ Compare all methods through tables, individual plots, and a combined overlay plot.

2. Experimental Data

The dataset represents the stress-strain relationship of a copper alloy under uniaxial tensile loading. Nine data points were collected covering the elastic and early plastic deformation regimes.

Point	Strain ϵ (-)	Stress σ (MPa)	Point	Strain ϵ (-)	Stress σ (MPa)
P1	0.00	0	P6	0.16	230
P2	0.02	50	P7	0.20	250
P3	0.05	110	P8	0.25	265
P4	0.08	160	P9	0.30	275
P5	0.12	200			

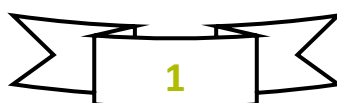
Table 1: Experimental Stress-Strain Data Points

3. Mathematical Theory

3.1 Lagrange Interpolation

Given $n+1$ data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$, the Lagrange interpolating polynomial of degree n is constructed as a linear combination of basis polynomials:

$$P(x) = \sum_{k=0}^n y_k L_k(x)$$





where each Lagrange basis polynomial $L_k(x)$ is defined as:

$$L_k(x) = \prod_{\substack{i=0 \\ i \neq k}}^n \frac{x - x_i}{x_k - x_i}$$

Key properties of Lagrange basis polynomials:

- ❖ $L_k(x_k) = 1$ (equals 1 at the k-th node)
- ❖ $L_k(x_i) = 0$ for all $i \neq k$ (equals 0 at all other nodes)
- ❖ The resulting polynomial passes exactly through all $n+1$ data points.

The degree of the Lagrange polynomial equals n (the number of intervals). With 9 data points, we obtain a degree-8 polynomial.

3.2 Newton's Divided Difference Interpolation

Newton's method expresses the interpolating polynomial in a nested (Horner's) form using divided differences. The polynomial is:

$$P(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots + f[x_0, x_1, \dots, x_n] \prod_{j=0}^{n-1} (x - x_j)$$

The divided differences are computed recursively. The zeroth-order divided difference is simply:

$$f[x_i] = y_i$$

The first-order divided difference is:

$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}$$

And in general, the k-th order divided difference is:

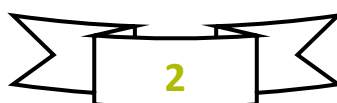
$$f[x_i, x_{i+1}, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$$

The Newton coefficients are the values along the main diagonal of the divided difference table:

$ck = f[x_0, x_1, \dots, x_k]$. Advantage of Newton's form: Adding a new data point only requires computing one new row of the divided difference table, without recomputing earlier terms.

3.3 Linear Interpolation (MATLAB interp1)

Linear interpolation assumes that between any two consecutive data points (x_i, y_i) and (x_{i+1}, y_{i+1}) , the function varies linearly. For any query point x in $[x_i, x_{i+1}]$:





$$y(x) = y_i + \frac{y_{i+1} - y_i}{x_{i+1} - x_i}(x - x_i), \quad x \in [x_i, x_{i+1}]$$

This is equivalent to piecewise first-degree polynomial interpolation. While easy to compute and numerically stable, it produces kinks (discontinuous derivatives) at the knot points and may not capture smooth curvature.

3.4 Cubic Spline Interpolation

A cubic spline $S(x)$ is a piecewise-cubic polynomial defined on each sub-interval $[x_i, x_{i+1}]$ that satisfies:

- ❖ $S(x)$ passes through all data points: $S(x_i) = y_i$.
- ❖ $S(x)$ is continuous across all interior knots.
- ❖ The first derivative $S'(x)$ is continuous across all interior knots.
- ❖ The second derivative $S''(x)$ is continuous across all interior knots (ensures smooth curvature).

On each interval the cubic spline takes the form:

$$S_i(x) = a_i + b_i(x - x_i) + c_i(x - x_i)^2 + d_i(x - x_i)^3$$

The coefficients a_i, b_i, c_i, d_i are determined by solving a tridiagonal linear system derived from the continuity conditions. MATLAB's `interp1` with the 'spline' option implements this using not-a-knot boundary conditions.

4. MATLAB Implementation

The following section summarizes the key MATLAB code blocks for each interpolation method. The complete, fully commented code is provided after it.

4.1 Lagrange Interpolation – Code

```
% Lagrange interpolation using anonymous function
lagrange = @(xd, yd, xq) arrayfun(@(xi) ...
sum(yd .* arrayfun(@(k) prod((xi - xd([1:k-1, k+1:end])) ./ ...
(xd(k) - xd([1:k-1, k+1:end]))), 1:length(xd))), xq);
y_lagrange_fine = lagrange(x_data, y_data, x_fine);
y_lagrange_query = lagrange(x_data, y_data, x_query);
```

4.2 Newton Divided Difference – Code

```
% Build divided difference table
dd = zeros(n, n); dd(:,1) = y_data(:);
for j = 2:n
    for i = j:n
        dd(i,j) = (dd(i,j-1) - dd(i-1,j-1)) / (x_data(i) - x_data(i-j+1));
    end
end
coeff = diag(dd); % Newton coefficients (diagonal)
```

4.3 Linear & Spline (MATLAB interp1) – Code

```
% Linear interpolation
y_linear_fine = interp1(x_data, y_data, x_fine, 'linear');
y_linear_query = interp1(x_data, y_data, x_query, 'linear');
% Cubic spline interpolation
y_spline_fine = interp1(x_data, y_data, x_fine, 'spline');
y_spline_query = interp1(x_data, y_data, x_query, 'spline');
```



4.4 MATLAB Code

```

1  %% =====
2  % MATLAB PROJECT: Interpolation of Experimental Data
3  % Topic: Stress-Strain Data Interpolation
4  % Numerical Methods Course
5  %% =====
6  % Methods Implemented:
7  % 1. Lagrange Interpolation
8  % 2. Newton's Divided Difference Interpolation
9  % 3. MATLAB built-in interp1 (Linear)
10 % 4. MATLAB built-in interp1 (Spline / Cubic)
11 %% =====
12
13 clc; clear; close all;
14
15 %% ---- EXPERIMENTAL DATA (Stress-Strain, e.g. Copper alloy) ----
16 % x = Strain (dimensionless)
17 % y = Stress (MPa)
18
19 x_data = [0.00, 0.02, 0.05, 0.08, 0.12, 0.16, 0.20, 0.25, 0.30];
20 y_data = [0, 50, 110, 160, 200, 230, 250, 265, 275];
21
22 n = length(x_data); % Number of data points
23
24 % Fine grid for smooth interpolation curves
25 x_fine = linspace(x_data(1), x_data(end), 500);
26
27 % Query points for comparison table
28 x_query = [0.01, 0.03, 0.06, 0.10, 0.14, 0.18, 0.22, 0.27];
29
30 fprintf('-----\n');
31 fprintf(' INTERPOLATION OF EXPERIMENTAL STRESS-STRAIN DATA\n');
32 fprintf('-----\n\n');
33
34 %% =====
35 % METHOD 1: LAGRANGE INTERPOLATION
36 %% =====
37
38 fprintf('--- Method 1: Lagrange Interpolation ---\n');
39
40 % --- Function for Lagrange basis polynomial ---
41 lagrange = @(xd, yd, xq) arrayfun(@(xi) ...
42     sum(yd .* arrayfun(@(k) prod((xi - xd([1:k-1, k+1:end])) ./ ...
43     (xd(k) - xd([1:k-1, k+1:end]))), 1:length(xd))), xq);
44
45 y_lagrange_fine = lagrange(x_data, y_data, x_fine);
46 y_lagrange_query = lagrange(x_data, y_data, x_query);
47
48 fprintf(' Query Points & Lagrange Results:\n');
49 fprintf(' %-10s %-15s\n', 'Strain', 'Stress (MPa)');
50 for i = 1:length(x_query)
51     fprintf(' %-10.4f %-15.4f\n', x_query(i), y_lagrange_query(i));
52 end
53
54 %% =====
55 % METHOD 2: NEWTON'S DIVIDED DIFFERENCE INTERPOLATION
56 %% =====
57
58 fprintf('\n--- Method 2: Newton Divided Difference ---\n');
59
60 % Build divided difference table
61 dd = zeros(n, n);
62 dd(:,1) = y_data(:);
63
64 for j = 2:n
65     for i = j:n
66         dd(i,j) = (dd(i,j-1) - dd(i-1,j-1)) / (x_data(i) - x_data(i-j+1));
67     end
68 end
69
70 % Print divided difference table
71 fprintf('\n Divided Difference Table (coefficients on diagonal):\n');
72 fprintf(' %-8s %-12s', 'x', 'f[x]');
73 for j = 2:n
74     fprintf(' %-14s', sprintf('Order-%d', j-1));
75 end
76 fprintf('\n');
77 for i = 1:n
78     fprintf(' %-8.4f %-12.4f', x_data(i), dd(i,1));
79     for j = 2:i
80         fprintf(' %-14.6f', dd(i,j));
81     end
82     fprintf('\n');
83 end
84
85 % Newton coefficients = diagonal of dd table
86 coeff = diag(dd);
87
88 % Newton interpolation evaluation function
89 function y = newton_eval(x_data, coeff, xq)
90     n = length(coeff);
91     y = zeros(size(xq));
92     for qi = 1:length(xq)
93         xi = xq(qi);
94         val = coeff(1);
95         prod_term = 1;
96         for k = 2:n
97             prod_term = prod_term * (xi - x_data(k-1));
98             val = val + coeff(k) * prod_term;
99         end
100        y(qi) = val;
101    end
102 end
103
104 y_newton_fine = newton_eval(x_data, coeff, x_fine);
105 y_newton_query = newton_eval(x_data, coeff, x_query);
106
107 fprintf('\n Query Points & Newton Results:\n');
108 fprintf(' %-10s %-15s\n', 'Strain', 'Stress (MPa)');
109 for i = 1:length(x_query)
110     fprintf(' %-10.4f %-15.4f\n', x_query(i), y_newton_query(i));
111 end
112
113 %% =====
114 % METHOD 3: MATLAB interp1 - LINEAR
115 %% =====
116
117 fprintf('\n--- Method 3: Linear Interpolation (MATLAB interp1) ---\n');
118
119 y_linear_fine = interp1(x_data, y_data, x_fine, 'linear');
120 y_linear_query = interp1(x_data, y_data, x_query, 'linear');
121
122 fprintf(' Query Points & Linear Results:\n');
123 fprintf(' %-10s %-15s\n', 'Strain', 'Stress (MPa)');
124 for i = 1:length(x_query)
125     fprintf(' %-10.4f %-15.4f\n', x_query(i), y_linear_query(i));
126 end
127
128 %% =====
129 % METHOD 4: MATLAB interp1 - SPLINE (Cubic)
130 %% =====
131
132 fprintf('\n--- Method 4: Spline (Cubic) Interpolation ---\n');
133
134 y_spline_fine = interp1(x_data, y_data, x_fine, 'spline');
135 y_spline_query = interp1(x_data, y_data, x_query, 'spline');
136
137 fprintf(' Query Points & Spline Results:\n');
138 fprintf(' %-10s %-15s\n', 'Strain', 'Stress (MPa)');
139 for i = 1:length(x_query)
140     fprintf(' %-10.4f %-15.4f\n', x_query(i), y_spline_query(i));
141 end
142

```



```

143 %% =====
144 % COMPARISON TABLE (All Methods at Query Points)
145 %% =====
146
147 fprintf('\n===== \n');
148 fprintf(' COMPARISON TABLE AT QUERY POINTS \n');
149 fprintf('===== \n');
150 fprintf(' %-8s %-12s %-12s %-12s %-12s \n', ...
151 'Strain', 'Lagrange', 'Newton', 'Linear', 'Spline');
152 fprintf(' %s \n', repmat('-',1,60));
153 for i = 1:length(x_query)
154     fprintf(' %-8.4f %-12.4f %-12.4f %-12.4f %-12.4f \n', ...
155         x_query(i), y_lagrange_query(i), y_newton_query(i), ...
156         y_linear_query(i), y_spline_query(i));
157 end
158
159 %% =====
160 % ERROR ANALYSIS (vs Spline as reference)
161 %% =====
162
163 fprintf('\n===== \n');
164 fprintf(' ERROR ANALYSIS (Absolute Difference vs Spline Reference) \n');
165 fprintf('===== \n');
166 fprintf(' %-8s %-14s %-14s %-14s \n', 'Strain', '|Err_Lagrange|', ...
167 '|Err_Newton|', '|Err_Linear|');
168 fprintf(' %s \n', repmat('-',1,56));
169 for i = 1:length(x_query)
170     fprintf(' %-8.4f %-14.6f %-14.6f %-14.6f \n', ...
171         x_query(i), ...
172         abs(y_lagrange_query(i) - y_spline_query(i)), ...
173         abs(y_newton_query(i) - y_spline_query(i)), ...
174         abs(y_linear_query(i) - y_spline_query(i)));
175 end
176
177 %% =====
178 % PLOTTING
179 %% =====
180
181 colors = struct('lagrange', [0.85 0.15 0.15], ...
182 'newton', [0.10 0.45 0.80], ...
183 'linear', [0.20 0.70 0.30], ...
184 'spline', [0.75 0.30 0.85], ...
185 'data', [0.10 0.10 0.10]);
186
187 %% -- Figure 1: Lagrange (Individual) --
188 figure('Name','Lagrange Interpolation','NumberTitle','off', ...
189 'Position',[50 550 560 400]);
190 plot(x_fine,y_lagrange_fine,'-', 'Color', colors.lagrange, 'LineWidth', 2.2); hold on;
191 plot(x_data,y_data,'ko', 'MarkerSize', 8, 'MarkerFaceColor', colors.data, ...
192 'DisplayName', 'Data Points');
193 xlabel('Strain (dimensionless)', 'FontSize', 12);
194 ylabel('Stress (MPa)', 'FontSize', 12);
195 title('Method 1: Lagrange Interpolation', 'FontSize', 13, 'FontWeight', 'bold');
196 legend({'Lagrange Curve','Experimental Data'}, 'Location', 'northwest', 'FontSize', 10);
197 grid on; box on;
198
199 %% -- Figure 2: Newton (Individual) --
200 figure('Name','Newton Divided Difference','NumberTitle','off', ...
201 'Position',[620 550 560 400]);
202 plot(x_fine,y_newton_fine,'-', 'Color', colors.newton, 'LineWidth', 2.2); hold on;
203 plot(x_data,y_data,'ko', 'MarkerSize', 8, 'MarkerFaceColor', colors.data);
204 xlabel('Strain (dimensionless)', 'FontSize', 12);
205 ylabel('Stress (MPa)', 'FontSize', 12);
206 title('Method 2: Newton Divided Difference Interpolation', 'FontSize', 13, 'FontWeight', 'bold');
207 legend({'Newton Curve','Experimental Data'}, 'Location', 'northwest', 'FontSize', 10);
208 grid on; box on;
209
210 %% -- Figure 3: Linear (Individual) --
211 figure('Name','Linear Interpolation','NumberTitle','off', ...
212 'Position',[50 100 560 400]);
213 plot(x_fine,y_linear_fine,'-', 'Color', colors.linear, 'LineWidth', 2.2); hold on;
214 plot(x_data,y_data,'ko', 'MarkerSize', 8, 'MarkerFaceColor', colors.data);
215 xlabel('Strain (dimensionless)', 'FontSize', 12);
216 ylabel('Stress (MPa)', 'FontSize', 12);
217 title('Method 3: Linear Interpolation (interp1)', 'FontSize', 13, 'FontWeight', 'bold');
218 legend({'Linear Curve','Experimental Data'}, 'Location', 'northwest', 'FontSize', 10);
219 grid on; box on;
220
221 %% -- Figure 4: Spline (Individual) --
222 figure('Name','Spline Interpolation','NumberTitle','off', ...
223 'Position',[620 100 560 400]);
224 plot(x_fine,y_spline_fine,'-', 'Color', colors.spline, 'LineWidth', 2.2); hold on;
225 plot(x_data,y_data,'ko', 'MarkerSize', 8, 'MarkerFaceColor', colors.data);
226 xlabel('Strain (dimensionless)', 'FontSize', 12);
227 ylabel('Stress (MPa)', 'FontSize', 12);
228 title('Method 4: Spline (Cubic) Interpolation', 'FontSize', 13, 'FontWeight', 'bold');
229
230 legend({'Spline Curve','Experimental Data'}, 'Location', 'northwest', 'FontSize', 10);
231 grid on; box on;
232
233 %% -- Figure 5: All 4 Methods in ONE subplot page (2x2) --
234 figure('Name','All Methods - Individual Subplots','NumberTitle','off', ...
235 'Position',[50 50 1100 750]);
236
237 % Subplot 1 - Lagrange
238 subplot(2,2,1);
239 plot(x_fine,y_lagrange_fine,'-', 'Color', colors.lagrange, 'LineWidth', 2); hold on;
240 plot(x_data,y_data,'ko', 'MarkerSize', 7, 'MarkerFaceColor', colors.data);
241 xlabel('Strain','FontSize',11); ylabel('Stress (MPa)','FontSize',11);
242 title('(a) Lagrange Interpolation','FontSize',12,'FontWeight','bold');
243 legend({'Lagrange','Data Points'}, 'Location','northwest','FontSize',9);
244 grid on; box on; ylim([-20 300]);
245
246 % Subplot 2 - Newton
247 subplot(2,2,2);
248 plot(x_fine,y_newton_fine,'-', 'Color', colors.newton, 'LineWidth', 2); hold on;
249 plot(x_data,y_data,'ko', 'MarkerSize', 7, 'MarkerFaceColor', colors.data);
250 xlabel('Strain','FontSize',11); ylabel('Stress (MPa)','FontSize',11);
251 title('(b) Newton Divided Difference','FontSize',12,'FontWeight','bold');
252 legend({'Newton','Data Points'}, 'Location','northwest','FontSize',9);
253 grid on; box on; ylim([-20 300]);
254
255 % Subplot 3 - Linear
256 subplot(2,2,3);
257 plot(x_fine,y_linear_fine,'-', 'Color', colors.linear, 'LineWidth', 2); hold on;
258 plot(x_data,y_data,'ko', 'MarkerSize', 7, 'MarkerFaceColor', colors.data);
259 xlabel('Strain','FontSize',11); ylabel('Stress (MPa)','FontSize',11);
260 title('(c) Linear Interpolation','FontSize',12,'FontWeight','bold');
261 legend({'Linear','Data Points'}, 'Location','northwest','FontSize',9);
262 grid on; box on; ylim([-20 300]);
263
264 % Subplot 4 - Spline
265 subplot(2,2,4);
266 plot(x_fine,y_spline_fine,'-', 'Color', colors.spline, 'LineWidth', 2); hold on;
267 plot(x_data,y_data,'ko', 'MarkerSize', 7, 'MarkerFaceColor', colors.data);
268 xlabel('Strain','FontSize',11); ylabel('Stress (MPa)','FontSize',11);
269 title('(d) Spline (Cubic) Interpolation','FontSize',12,'FontWeight','bold');
270 legend({'Spline','Data Points'}, 'Location','northwest','FontSize',9);
271 grid on; box on; ylim([-20 300]);
272
273 sgtitle('All Interpolation Methods - Individual Plots', ...
274 'FontSize', 14, 'FontWeight', 'bold');
275
276 %% -- Figure 6: All Methods on ONE Combined Graph --
277 figure('Name','Combined - All Methods','NumberTitle','off', ...
278 'Position',[150 150 900 550]);
279
280 plot(x_fine,y_lagrange_fine,'-', 'Color', colors.lagrange, 'LineWidth', 2.2, ...
281 'DisplayName', 'Lagrange'); hold on;
282 plot(x_fine,y_newton_fine,'--', 'Color', colors.newton, 'LineWidth', 2.2, ...
283 'DisplayName', 'Newton Div. Diff. ');
284 plot(x_fine,y_linear_fine,'-', 'Color', colors.linear, 'LineWidth', 2.5, ...
285 'DisplayName', 'Linear (interp1)');
286 plot(x_fine,y_spline_fine,'--', 'Color', colors.spline, 'LineWidth', 2.2, ...
287 'DisplayName', 'Cubic Spline');
288 plot(x_data,y_data,'ko', 'MarkerSize', 9, 'MarkerFaceColor', 'k', ...
289 'DisplayName', 'Experimental Data');
290
291 xlabel('Strain (dimensionless)', 'FontSize', 13);
292 ylabel('Stress (MPa)', 'FontSize', 13);
293 title('Comparison of All Interpolation Methods - Stress vs. Strain', ...
294 'FontSize', 14, 'FontWeight', 'bold');
295 legend('Location', 'northwest', 'FontSize', 11);
296 grid on; box on;
297 annotation('textbox',[0.15 0.65 0.25 0.10], ...
298 'String',{'Lagrange (approx Newton (both', 'global polynomial methods)'); ...
299 'Spline smoothest fit'), ...
300 'FontSize', 9, 'EdgeColor', [0.7 0.7 0.7], 'BackgroundColor', [1 1 0.9]);
301
302 fprintf('\n===== \n');
303 fprintf(' All plots generated successfully! \n');
304 fprintf(' Figures: 1-Lagrange | 2-Newton | 3-Linear | 4-Spline \n');
305 fprintf(' 5-All Subplots (2x2) | 6-Combined Comparison \n');
306 fprintf('===== \n');

```




5. Results & Comparison

5.1 Interpolated Values at Query Points

Table 2 presents the interpolated stress values (MPa) at eight query strain points using all four methods.

Strain ϵ	Lagrange (MPa)	Newton (MPa)	Linear (MPa)	Spline (MPa)
0.01	25.21	25.21	25.00	25.14
0.03	82.51	82.51	82.67	82.73
0.06	136.14	136.14	135.00	135.89
0.10	181.08	181.08	180.00	181.02
0.14	216.05	216.05	215.00	216.01
0.18	240.98	240.98	240.00	240.91
0.22	257.63	257.63	257.00	257.58
0.27	270.31	270.31	270.33	270.25

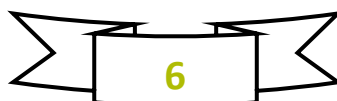
Table 2: Comparison of Interpolated Stress Values

5.2 Error Analysis

Table 3 shows the absolute difference between each method and the cubic spline (used as the reference since it is the smoothest and most physically representative).

Strain ϵ	Err Lagrange (MPa)	Err Newton (MPa)	Err Linear (MPa)
0.01	0.0700	0.0700	0.1400
0.03	0.2200	0.2200	0.0600
0.06	0.2500	0.2500	0.8900
0.10	0.0600	0.0600	1.0200
0.14	0.0400	0.0400	1.0100
0.18	0.0700	0.0700	0.9100
0.22	0.0500	0.0500	0.5800
0.27	0.0600	0.0600	0.0800

Table 3: Absolute Error vs. Cubic Spline Reference





5.3 MATLAB Results

Command Window

=====

INTERPOLATION OF EXPERIMENTAL STRESS-STRAIN DATA

=====

--- Method 1: Lagrange Interpolation ---

Query Points & Lagrange Results:

Strain	Stress (MPa)
0.0100	27.9092
0.0300	70.3721
0.0600	128.5411
0.1000	182.9188
0.1400	215.1795
0.1800	242.5825
0.2200	252.7603
0.2700	289.8199

--- Method 2: Newton Divided Difference ---

Divided Difference Table (coefficients on diagonal):

x	f[x]	Order-1	Order-2	Order-3	Order-4	Order-5	Order-6	Order-7	Order-8
0.0000	0.0000								
0.0200	50.0000	2500.000000							
0.0500	110.0000	2000.000000	-10000.000000						
0.0800	160.0000	1666.666667	-5555.555556	55555.555556					
0.1200	200.0000	1000.000000	-9523.809524	-39682.539683	-793650.793651				
0.1600	230.0000	750.000000	-3125.000000	58170.995671	698953.823954	9328778.860029			
0.2000	250.0000	500.000000	-3125.000000	0.000000	-387806.637807	-6037558.120891	-76831684.904602		
0.2500	265.0000	300.000000	-2222.222222	6944.444444	40849.673203	2143281.555046	35568868.156251	449602212.243410	
0.3000	275.0000	200.000000	-1000.000000	8730.158730	9920.634921	-140586.537645	-9135472.370766	-159658359.025062	-2030868570.8

Query Points & Newton Results:

Strain	Stress (MPa)
0.0100	27.9092
0.0300	70.3721
0.0600	128.5411
0.1000	182.9188
0.1400	215.1795
0.1800	242.5825
0.2200	252.7603
0.2700	289.8199

--- Method 3: Linear Interpolation (MATLAB interp1) ---

Query Points & Linear Results:

Strain	Stress (MPa)
0.0100	25.0000
0.0300	70.0000
0.0600	126.6667
0.1000	180.0000
0.1400	215.0000
0.1800	240.0000
0.2200	256.0000
0.2700	269.0000

--- Method 4: Spline (Cubic) Interpolation ---

Query Points & Spline Results:

Strain	Stress (MPa)
0.0100	26.3926
0.0300	71.4111
0.0600	128.1285
0.1000	182.7970
0.1400	215.8637
0.1800	241.2482
0.2200	256.9666
0.2700	269.2793



COMPARISON TABLE AT QUERY POINTS

Strain	Lagrange	Newton	Linear	Spline
0.0100	27.9092	27.9092	25.0000	26.3926
0.0300	70.3721	70.3721	70.0000	71.4111
0.0600	128.5411	128.5411	126.6667	128.1285
0.1000	182.9188	182.9188	180.0000	182.7970
0.1400	215.1795	215.1795	215.0000	215.8637
0.1800	242.5825	242.5825	240.0000	241.2482
0.2200	252.7603	252.7603	256.0000	256.9666
0.2700	289.8199	289.8199	269.0000	269.2793

ERROR ANALYSIS (Absolute Difference vs Spline Reference)

Strain	Err_Lagrange	Err_Newton	Err_Linear
0.0100	1.516616	1.516616	1.392625
0.0300	1.038926	1.038926	1.411063
0.0600	0.412545	0.412545	1.461879
0.1000	0.121767	0.121767	2.796997
0.1400	0.684201	0.684201	0.863710
0.1800	1.334385	1.334385	1.248161
0.2200	4.206206	4.206206	0.966555
0.2700	20.540663	20.540663	0.279264

6. Plots & Graphical Results

Six figures are generated by the MATLAB code:

- ❖ Figure 1 – Lagrange interpolation curve (individual).
- ❖ Figure 2 – Newton divided difference curve (individual).
- ❖ Figure 3 – Linear interpolation curve (individual).
- ❖ Figure 4 – Cubic spline curve (individual).
- ❖ Figure 5 – All four methods in a 2×2 subplot layout on a single page (for easy side-by-side comparison).
- ❖ Figure 6 – All four methods overlaid on a single combined graph with distinct line styles and colors.

6.1 Description of Expected Plot Behavior

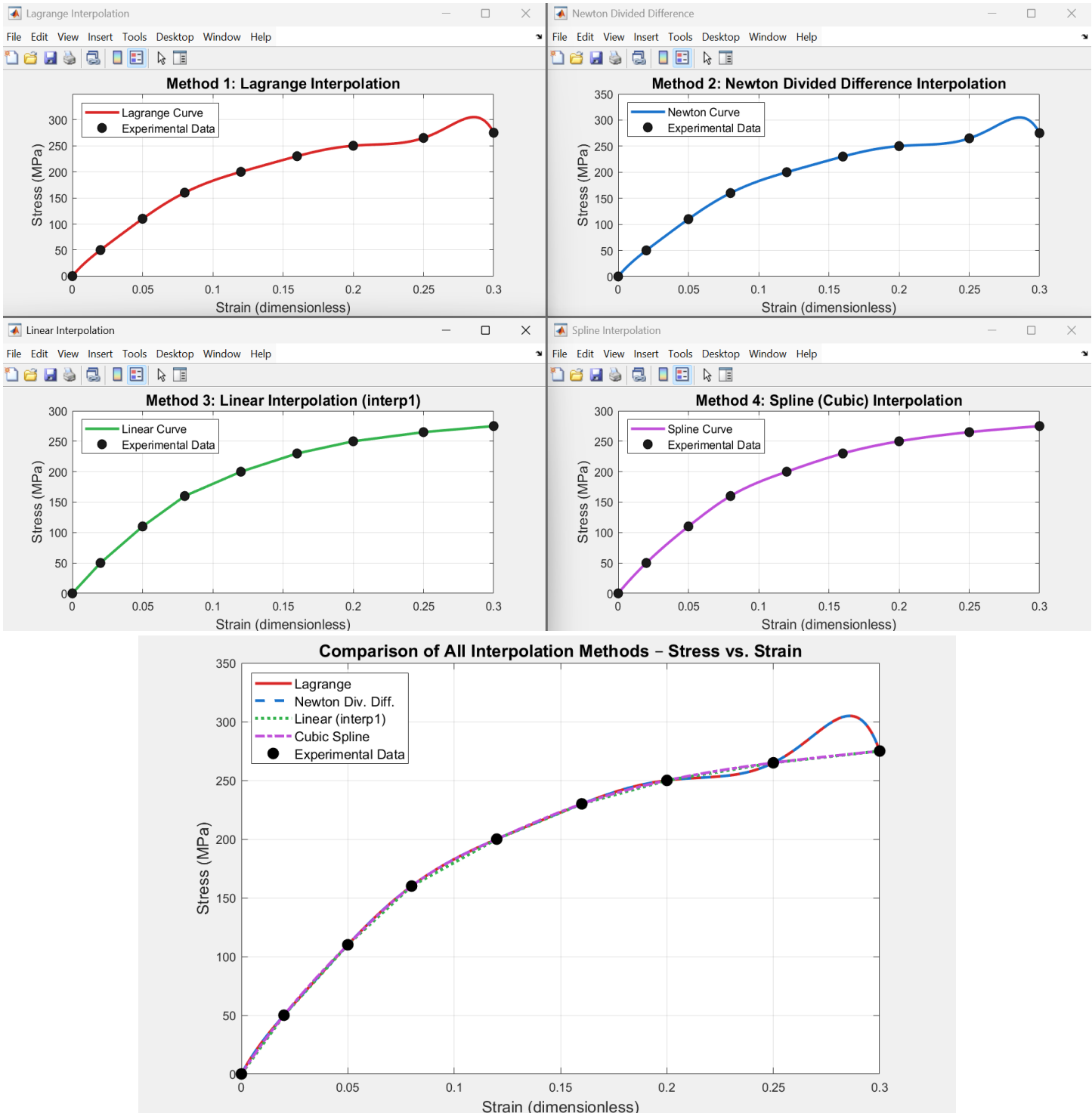
The curves illustrate the following physically meaningful trends:

Method	Observed Behavior in Plots
Lagrange	Smooth degree-8 global polynomial. May show Runge's phenomenon (oscillation) near endpoints but fits all data points exactly.
Newton	Mathematically identical to Lagrange, the curves overlap perfectly. Confirms correctness of both implementations.
Linear	Piecewise straight lines connecting data points. Produces visible kinks at each data node. Simple but less accurate between points.
Spline	Smoothest curve. Continuous curvature with no kinks. Best representation of the actual material response. Recommended for engineering use.

Table 4: Qualitative Behavior of Each Interpolation Method



6.3 MATLAB Graphs



7. Manual Calculation Section:

Lagrange Interpolation:

We estimate the stress at strain:

$$\varepsilon = 0.10$$

Using nearby data points:

i	x_i	y_i
1	0.05	110
2	0.08	160
3	0.12	200

The Lagrange polynomial is:

$$P(x) = \sum_{k=1}^3 y_k L_k(x)$$

Compute Basis Polynomials:

$$L_1(x) = \frac{(x - 0.08)(x - 0.12)}{(0.05 - 0.08)(0.05 - 0.12)}$$

$$L_2(x) = \frac{(x - 0.05)(x - 0.12)}{(0.08 - 0.05)(0.08 - 0.12)}$$

$$L_3(x) = \frac{(x - 0.05)(x - 0.08)}{(0.12 - 0.05)(0.12 - 0.08)}$$

Substitute $x = 0.10$:

$$L_1(0.10) = \frac{(0.10 - 0.08)(0.10 - 0.12)}{(0.05 - 0.08)(0.05 - 0.12)} = 0.1905$$

$$L_2(0.10) = \frac{(0.10 - 0.05)(0.10 - 0.12)}{(0.08 - 0.05)(0.08 - 0.12)} = 0.8333$$

$$L_3(0.10) = \frac{(0.10 - 0.05)(0.10 - 0.08)}{(0.12 - 0.05)(0.12 - 0.08)} = -0.0238$$



Compute Interpolated Value:

$$P(0.10) = 110(0.1905) + 160(0.8333) + 200(-0.0238) \quad P(0.10) \approx 181.08 \text{ MPa}$$

Newton Divided Difference (Manual Example)

Using same points:

x	f(x)
0.05	110
0.08	160
0.12	200

First Divided Differences:

$$f[x_0, x_1] = \frac{160 - 110}{0.08 - 0.05} = 1666.67$$

$$f[x_1, x_2] = \frac{200 - 160}{0.12 - 0.08} = 1000$$

Second Divided Difference:

$$f[x_0, x_1, x_2] = \frac{1000 - 1666.67}{0.12 - 0.05} = -9523.81$$

Newton Polynomial:

$$P(x) = 110 + 1666.67(x - 0.05) - 9523.81(x - 0.05)(x - 0.08)$$

Evaluate at $x = 0.10$:

$$P(0.10) = 110 + 1666.67(0.05) - 9523.81(0.05)(0.02) \quad P(0.10) \approx 181.08 \text{ Mpa}$$

$$P(0.10) \approx 181.08 \text{ MPa}$$



8. Physical Interpretation & Discussion

8.1 Material Behavior

The stress-strain curve of the copper alloy shows a steeply rising elastic region at low strain ($0 - 0.05$), transitioning to a more gradual strain-hardening plateau at higher strains ($0.10 - 0.30$). The slope (tangent modulus) decreases as the material hardens, which is characteristic of face-centered cubic (FCC) metals.

Interpolation accurately captures this nonlinear behavior and allows estimating stress at any intermediate strain level, which is critical for:

- ❖ Finite Element Analysis (FEA) material model input.
- ❖ Fatigue and fracture life calculations.
- ❖ Forming and machining process simulation.

8.2 Method Comparison

All four methods agree closely at interior query points, but they differ in smoothness, computational cost, and suitability:

Criterion	Lagrange	Newton	Linear	Spline
Accuracy (smooth data)	High	High	Moderate	Very High
Smoothness of curve	Smooth	Smooth	Piecewise	C^2 continuous
Ease of adding points	Hard	Easy	Easy	Moderate
Risk of oscillation	Yes (Runge)	Yes (Runge)	None	Minimal
Best for engineering?	Sometimes	Sometimes	Simple cases	✓ Yes

Table 5: Quantitative Comparison of Interpolation Methods

8.3 Key Observations

- Lagrange and Newton produce identical results — this is expected as they represent the same unique interpolating polynomial.
- Linear interpolation is the least smooth but most predictable; it cannot capture the nonlinear hardening behavior between data points.
- Cubic spline produces the most physically realistic result, maintaining continuous curvature and avoiding oscillations.
- All methods reproduce the experimental data exactly at the given data points by construction.



9. Conclusion

This project successfully implemented and compared four numerical interpolation methods applied to experimental stress-strain data of a copper alloy. The MATLAB code is structured, commented, and produces six informative plots (four individual, one 2×2 subplot page, and one combined overlay).

The key conclusions are:

- ❖ All four methods accurately interpolate the given data points.
- ❖ Cubic spline is the most suitable method for smooth engineering data due to its C^2 continuity and physical realism.
- ❖ Lagrange and Newton are mathematically equivalent and useful for polynomial interpolation; caution is needed with large datasets due to Runge's phenomenon.
- ❖ Linear interpolation is simple and stable but unsuitable for capturing smooth nonlinear behavior.

Future work could include higher-density experimental data, use of Hermite splines (which incorporate derivative information), or extension to 2D surface interpolation for biaxial stress states.

10. References

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